SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 25**
Question Count: 5

Code of Conduct

I Amit Krishna (192240047) (Name / Reg. No.)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Amy

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

Total Marks:

1. Cloud computing is a technology that provides computing resources such as servers, storage, DB & software over the internet. They have the following characteristics

i) On-demand self service:

Users can obtain computing resources whenever required without any assistance

ii) Scalability:

Resources can be increased or decreased according to the workload or user requirements.

iii) Pay for usage:

Users generally pay only for the resources & services they actually use.

iv) Broad network access:

Cloud services can be accessed from anywhere using the internet & devices such as computers, tablets & phones

2. Cloud computing developed through several stages of technological evolution:

* Initially, mainframe computers provided centralized computing.
* This could later be done using personal computers & client-server systems.
* The development of virtualization allowed one server to run multiple virtual machines
* Parallely, the growth of Internet, high-speed networks, web services & distributed computing made it possible to access computing resources remotely
* These developments lead to modern cloud platforms such as AWS, Microsoft Azure, & google cloud
* Now, computing resources & computing services are provided over the internet & they can be accessed easily & remotely.

3. Server virtualization is the process of dividing a physical server into multiple virtual servers using a software layer

The software layer is called a hypervisor & it enables each server to run its own operating system & applications.

Virtualization helps cloud providers use hardware efficiently, reducing costs & quickly create or remove virtual machines.

Applications of virtualization are as follows:

* Virtualization as an enabling technology:
  It is a technology used to efficiently manage & share physical computing resources.
* Cloud computing as a service model:
  It provides computing resources to users over the internet, usually on-demand & with usage-based pricing.